1 A student is investigating the stability of a wooden block resting on a bench.

A strip is attached by a nail to the centre of the top of the block and is able to rotate, as shown in Fig. 1.1 and Fig. 1.2.

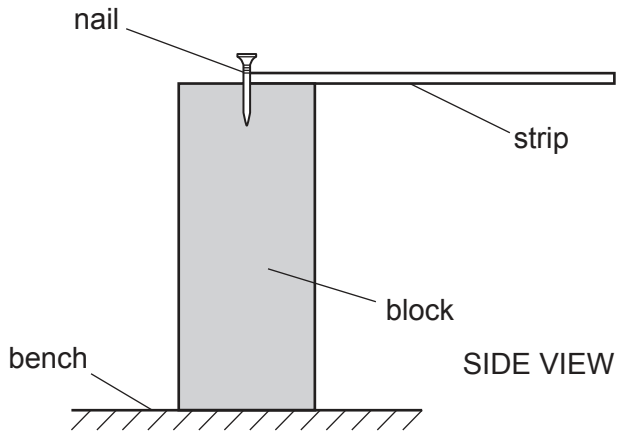

Fig. 1.1

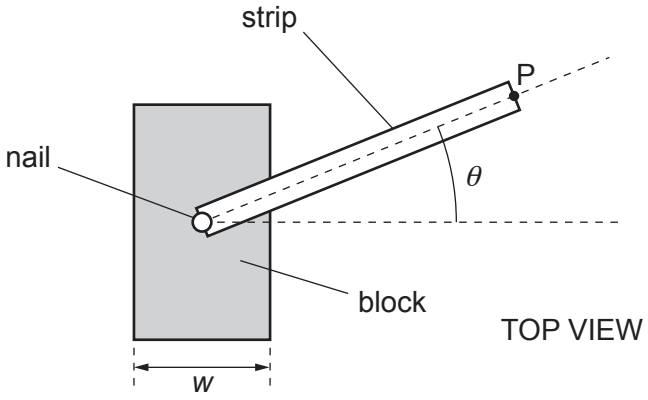

Fig. 1.2

A load of mass m is attached to the free end of the strip at point P. The student is investigating the position of the strip indicated by angle θ, as shown in Fig. 1.2, at which the block just topples.

It is suggested that the relationship between m and θ is

$$\alpha V w = 2mL\cos\theta - mw$$

where α is a constant, V is the volume of the block, w is the width of the block and L is the distance between the centre of the nail and the centre of the load.

Design a laboratory experiment to test the relationship between m and θ.
Explain how your results could be used to determine a value for α.

You should draw a diagram, on page 3, showing the arrangement of your equipment.
In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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